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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Geometrical Optics

English Medium

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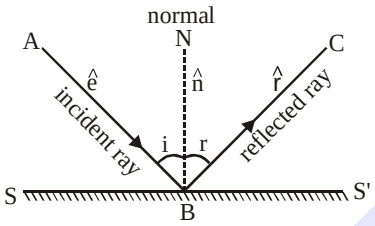
GEOMETRICAL OPTICS

KEY CONCEPT

1. LAWS OF REFLECTION

(i) The incident ray (AB), the reflected ray (BC) and normal (NB) to the surface (SS') of reflection at the point of incidence (B) lie in the same plane. This plane is called the plane of incidence (also plane of reflection).

(ii) The angle of incidence (the angle between normal and the incident ray) and the angle of reflection (the angle between the reflected ray and the normal) are equal $\boxed{\angle i = \angle r}$



In vector form $\boxed{\hat{r} = \hat{e} - 2(\hat{e}.\hat{n})\hat{n}}$

2. OBJECT

(a) **Real :** Point from which rays actually diverge.

(b) **Virtual:** Point towards which rays appear to converge

3. IMAGE :

Image is decided by reflected or refracted rays only. The point image for a mirror is that point

(i) Towards which the rays reflected from the mirror, actually converge (real image).

OR

(ii) From which the reflected rays appear to diverge (virtual image) .

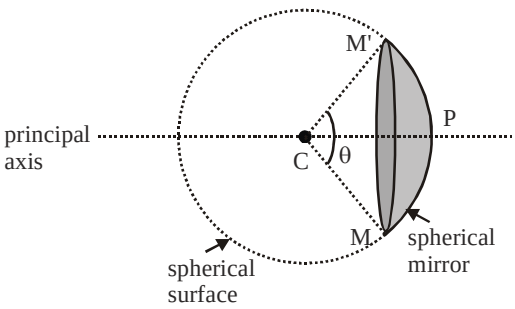
4. CHARACTERISTICS OF REFLECTION BY A PLANE MIRROR

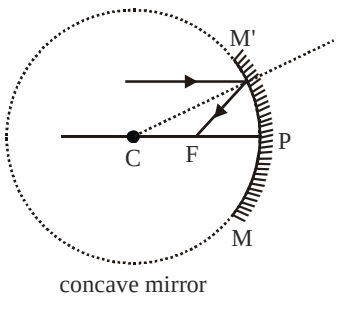
(a) The size of the image is the same as that of the object.

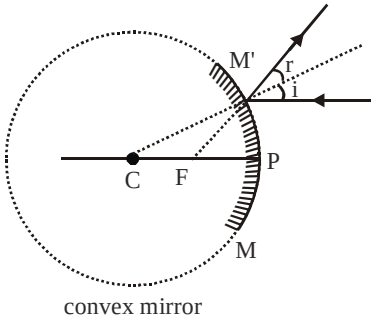
(b) For a real object the image is virtual and for a virtual object the image is real.

(c) For a fixed incident light ray, if the mirror be rotated through an angle θ the reflected ray turns through an angle 2θ.

5. SPHERICAL MIRRORS







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6. PARAXIAL RAYS

Rays which forms very small angle with axis are called paraxial rays.

7. SIGN CONVENTION

We follow Cartesian coordinate system convention according to which

(a) The pole of the mirror is the origin .

(b) The direction of the incident rays is considered as positive x-axis.

(c) Vertically up is positive y-axis.

Note : According to above convention radius of curvature and focus of concave mirror is negative and of convex mirror is positive.

8. MIRROR FORMULA : $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$.

f = x- coordinate of focus ; u = x-coordinate of object ; v = x-coordinate of image

Note : Valid only for paraxial rays.

9. TRANSVERSE MAGNIFICATION $m = \frac{h_2}{h_1} = -\frac{v}{u}$

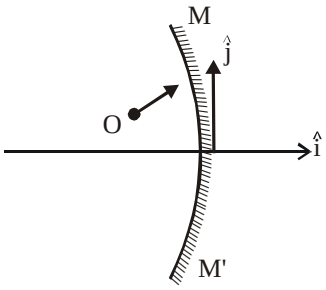
h_2 = y co-ordinate of image h_1 = y co-ordinate of the object

(both perpendicular to the principal axis of mirror)

10. VELOCITY OF IMAGE OF MOVING OBJECT (SPHERICAL MIRROR)

(a) Velocity component along axis (Longitudinal velocity)

When an object is coming from infinite towards the focus of concave mirror



$$\because \frac{1}{v}+\frac{1}{u}=\frac{1}{f} \quad \therefore -\frac{1}{v^2}\frac{dv}{dt}-\frac{1}{u^2}\frac{du}{dt}=0$$

$$\Rightarrow \vec{v}_{ix} = -\frac{v^2}{u^2}\vec{v}_{ox} = -m^2\vec{v}_{ox}$$

$$\left[v_{ix} = \frac{dv}{dt} = \text{velocity of image along principal-axis; } v_{ox} = \frac{du}{dt} = \text{velocity of object along principal-axis} \right]$$

(b) <u>Velocity component perpendicular to axis</u> (Transverse velocity)

$$m = \frac{h_I}{h_0} = -\frac{v}{u} = \frac{f}{f-u} \Rightarrow h_I = \left(\frac{f}{f-u}\right)h_0$$

$$\frac{dh_I}{dt} = \left(\frac{f}{f-u}\right)\frac{dh_0}{dt} + \frac{f\,h_0}{(f-u)^2}\frac{du}{dt} \Rightarrow \vec{v}_{iy} = \left[m\vec{v}_{oy} + \frac{m^2 h_0}{f}\vec{v}_{ox} \right]\hat{j}$$

$$\left[\begin{array}{l} \frac{dh_I}{dt} = \text{velocity of image } \perp^r \text{ to principal-axis} \\ \frac{dh_o}{dt} = \text{velocity of object } \perp^r \text{ to principal-axis} \end{array} \right]$$

Note : Here principal axis has been taken to be along x–axis.

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11. NEWTON'S FORMULA

Applicable to a pair of real object and real image position only. They are called conjugate positions or foci X_1, X_2 are the distance along the principal axis of the real object and real image respectively from the principal focus.

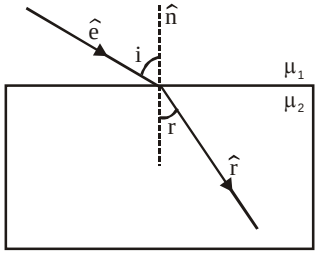
$$\boxed{X_1X_2 = f^2}$$

12. OPTICAL POWER

Optical power of a mirror (in Diopters) $= -\frac{1}{f}$; f = focal length (in meters) with sign.

13. LAWS OF REFRACTION (AT ANY REFRACTING SURFACE)

(i) Incident ray, refracted ray and normal always lie in the same plane.



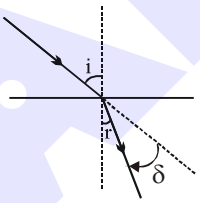
In vector form $\boxed{(\hat{e} \times \hat{n}).\hat{r} = 0}$

(ii) The product of refractive index and sine of angle of incidence at a point in a medium is constant. $\mu_1 \sin i = \mu_2 \sin r$ (Snell's law)

Snell's law $\boxed{\frac{\text{Sin i}}{\text{Sin r}} = {}_1n_2 = \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}}$ In vector form $\boxed{\mu_1|\hat{e} \times \hat{n}| = \mu_2|\hat{r} \times \hat{n}|}$

Note : Frequency of light does not change during refraction .

14. DEVIATION OF A RAY DUE TO REFRACTION



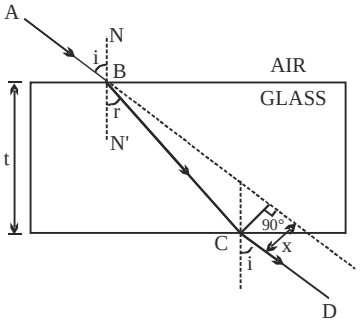
$\boxed{\text{angle of deviation, } \delta = i - r}$

15. REFRACTION THROUGH A PARALLEL SLAB :

(i) Emerged ray is parallel to the incident ray, if medium is same on both sides.

(ii) Lateral shift $x = \frac{t \sin(i - r)}{\cos r}$

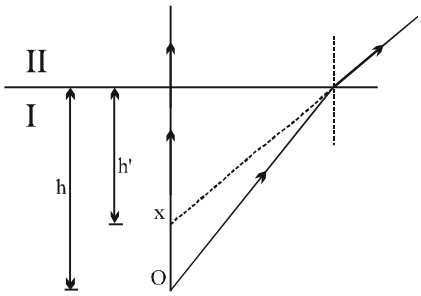
t = thickness of slab



Note : Emerged ray will not be parallel to the incident ray if the medium on both the sides are different

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16. APPARENT DEPTH OF SUBMERGED OBJECT

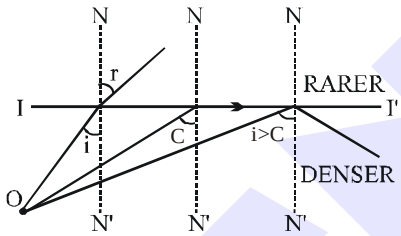


$(h' < h)$

For near normal incidence $h' = \frac{\mu_2}{\mu_1} h$

Note : h and h' are always measured from surface.

17. CRITICAL ANGLE & TOTAL INTERNAL REFLECTION (T. I. R.)



Conditions of T. I. R.

(i) Ray is going from denser to rarer medium

(ii) Angle of incidence should be greater than the critical angle (i > C) .

Critical angle $C = \sin^{-1}\frac{n_r}{n_i}$

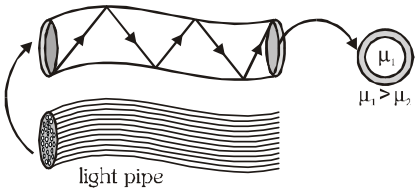
18. SOME ILLUSTRATIONS OF TOTAL INTERNAL REFLECTION

(i) Sparkling of diamond

The sparkling of diamond is due to total internal reflection inside it. As refractive index for diamond is 2.5 so C = 24°. Now the cutting of diamond are such that i > C. So TIR will take place again and again inside it. The light which beams out from a few places in some specific directions makes it sparkle.

(ii) Optical Fibre

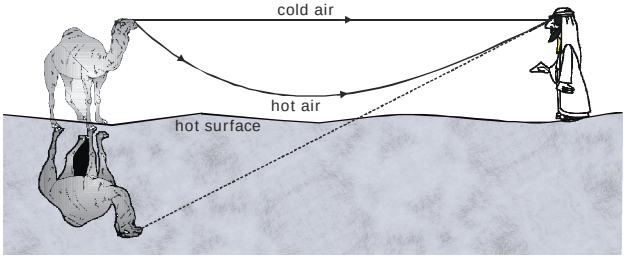
In it light through multiple total internal reflections is propagated along the axis of a glass fibre of radius of few microns in which index of refraction of core is greater than that of surroundings.

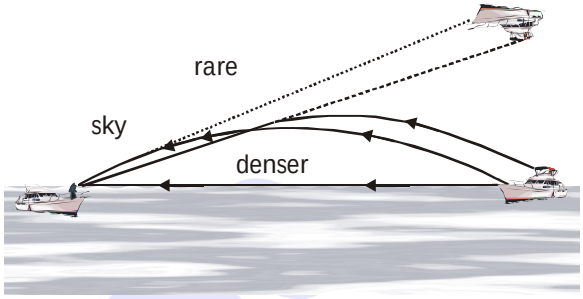


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(iii) Mirage and looming

Mirage is caused by total internal reflection in deserts where due to heating of the earth, refractive index of air near the surface of earth becomes lesser than above it. Light from distant objects reaches the surface of earth with $i > \theta_C$ so that TIR will take place and we see the image of an object along with the object as shown in figure.





Similar to 'mirage' in deserts, in polar regions 'looming' takes place due to TIR. Here μ decreases with height and so the image of an object is formed in air if (i>C) as shown in figure.

19. REFRACTION AT SPHERICAL SURFACE

(a) $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

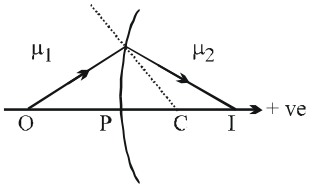
v, u & R are to be kept with sign

as v = PI

u = –PO

R = PC

(Note : Radius is with sign)



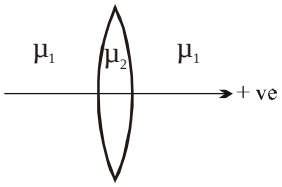
(b) $m = \frac{\mu_1 v}{\mu_2 u}$

20. IMAGE FORMATION DUE TO THIN LENS

(a) Lens' fomula $\Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

(b) Lens' maker fomula $\Rightarrow \frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

(c) magnification due to lens' $\Rightarrow m = \frac{v}{u}$



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21. POWER OF LENS

Reciprocal of focal length in meter is known as power of lens.

SI unit : dioptre (D) **Power of lens :** $P = \frac{\mu_1}{f(m)}$ dioptre

22. COMBINATION OF LENS

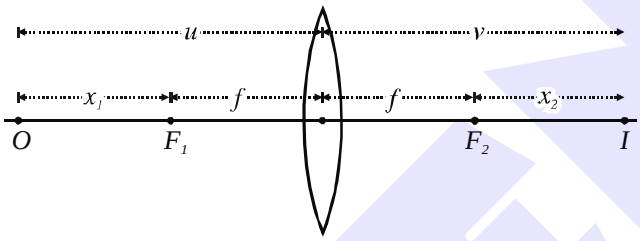
Two thin lens are placed in contact to each other



power of combination. $P = P_1 + P_2 \Rightarrow \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Use sign convention when solve numericals

23. NEWTON'S FORMULA FOR LENS



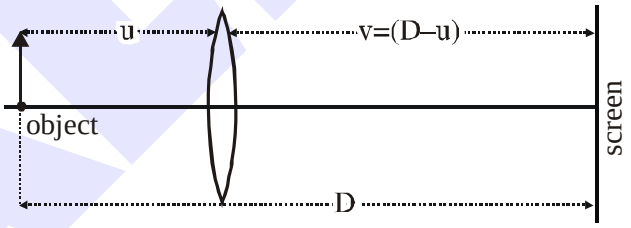
$f = \sqrt{x_1 x_2}$

x_1 = distance of object from focus; x_2 = distance of image from focus.

24. DISPLACEMENT METHOD FOR CONVEX LENS

It is used for determination of focal length of convex lens in laboratory.

A thin convex lens of focal length f is placed between an object and a screen fixed at a distance D apart.



If D > 4f there are two position of lens at which a sharp image of the object is formed on the screen

By lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ we have $\frac{1}{D-u} - \frac{1}{-u} = \frac{1}{f}$

$\Rightarrow u^2 - Du + Df = 0 \Rightarrow u = \frac{D \pm \sqrt{D(D-4f)}}{2}$ there are three possibilities

(i) **For D < 4f :** u will be imaginary hence physically no position of lens is possible

(ii) **For D = 4f :** $u = \frac{D}{2} = 2f$ so only one position of lens is possible

and since v = D – u = 4f – 2f = u =2f

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(iii) **For D > 4f :** $u_1 = \frac{D-\sqrt{D(D-4f)}}{2}$ and $u_2 = \frac{D+\sqrt{D(D-4f)}}{2}$

So there are two positions of lens for which real image will be formed on the screen.

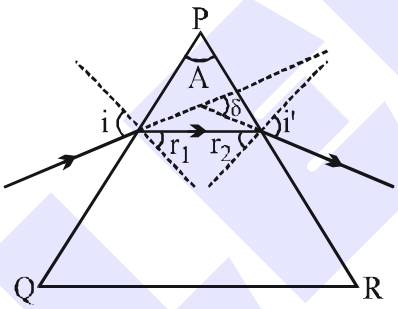
for two positions of the lens distances of object and image are interchangeable.

so $u_1 = \frac{D-x}{2} = v_2$ and $v_1 = \frac{D+x}{2} = u_2$

$m_1 = \frac{I_1}{O} = \frac{v_1}{u_1} = \frac{D+x}{D-x}$ and $m_2 = \frac{I_2}{O} = \frac{v_2}{u_2} = \frac{D-x}{D+x}$

Now $m_1 \times m_2 = \frac{D+x}{D-x} \times \frac{D-x}{D+x} \Rightarrow \frac{I_1 I_2}{O^2} = 1 \Rightarrow O = \sqrt{I_1 I_2}$

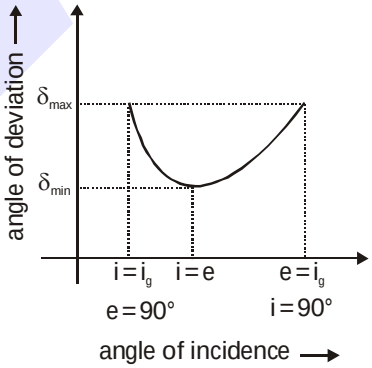
25. REFRACTION THROUGH PRISM



(i) $\delta = (i + i') - (r + r')$

(ii) $r + r' = A$

(iii) Variation of δ versus i (shown in diagram) .



(iv) There is one and only one angle of incidence for which the angle of deviation is minimum. When $\delta = \delta_m$ then i = i′ & r = r′, the ray passes symmetrically about the prism, & then

$n = \frac{\sin\left[\frac{A+\delta m}{2}\right]}{\sin\left[\frac{A}{2}\right]}$, where n = absolute R.I. of glass .

Note : When the prism is dipped in a medium then n = R.I. of glass w.r.t. medium.

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(v) For a thin prism ($A \leq 10^o$) ; $\delta = (n - 1) A$

(vi) **Dispersion Of Light :**

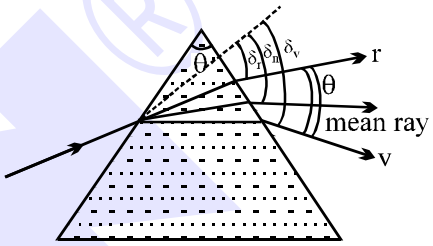
The angular splitting of a ray of white light into a number of components when it is refracted in a medium other than air is called ***Dispersion of Light.***

(vii) **Angle of Dispersion :**

Angle between the rays of the extreme colors in the refracted (dispersed) light is called **Angle of Dispersion.** $\theta = \delta_v - \delta_r$.

(viii) Dispersive power (ω) of the medium of the material of prism.

$$\omega = \frac{\text{angular dispersion}}{\text{deviation of mean ray (yellow)}}$$



For small angled prism ($A \leq 10^o$)

$$\omega = \frac{\delta_v - \delta_R}{\delta_y} = \frac{n_v - n_R}{n - 1} \; ; n = \frac{n_v + n_R}{2}$$

n_v, n_R & n are R. I. of material for violet, red & yellow colours respectively.

26. COMBINATION OF TWO PRISMS

(a) **Achromatic Combination (deviation without dispersion) :**

It is used for deviation without dispersion . Condition for this $(n_v - n_r) A = (n'_v - n'_r) A'$.

$$\text{Net mean deviation} = \left[\frac{n_v + n_R}{2} - 1\right] A - \left[\frac{n'_v + n'_R}{2} - 1\right] A' .$$

or $\omega\delta + \omega'\delta' = 0$ where ω, ω' are dispersive powers for the two prisms & δ, δ' are the mean deviation.

(b) **Direct Vision Combination (dispersion without deviation) :**

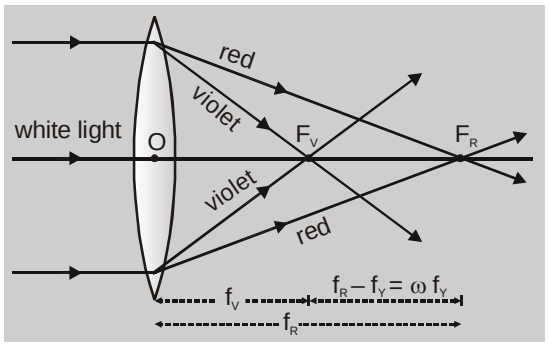
It is used for producing dispersion without deviation condition for this

$$\left[\frac{n_v + n_R}{2} - 1\right] A = \left[\frac{n'_v + n'_R}{2} - 1\right] A' .$$

Net angle of dispersion = $(n_v - n_r) A - (n_v' - n_r') A'$.

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27. CHROMATIC ABERRATION IN LENS



The image of a object in white light formed by a lens is usually coloured and blurred. This defect of image is called chromatic aberration and arises due to the fact that focal length of a lens is different for different colours. For a single lens $\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$ and as μ of lens is maximum for violet while minimum for red, violet is focused nearest to the lens while red farthest from it It is defect of lens.

28. ACHROMATIC COMBINATION OF LENS

If two or more lens combined together in such a way that this combination produce image at a same point then this combination is known as achromatic combination of lenses.

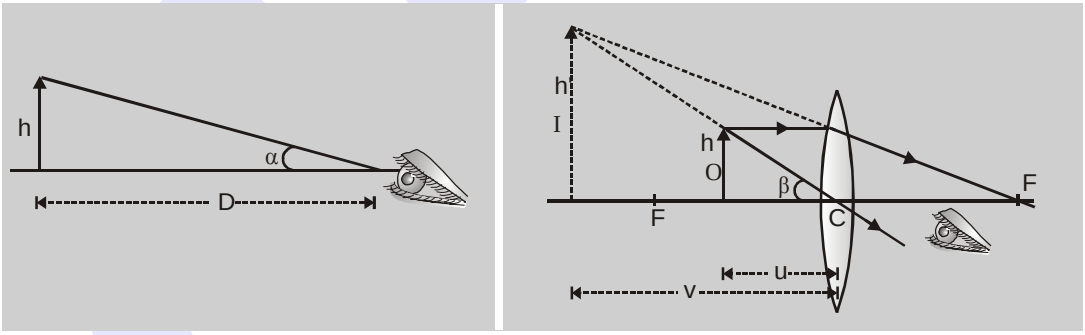
$$\frac{\omega}{f_y} + \frac{\omega'}{f'_y} = 0 \Rightarrow \frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0 \Rightarrow \frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2}$$

For combination of lens. $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$ (Apply sign convention in numerical)

29. OPTICAL INSTRUMENT

(a) Simple microscope

When object is placed between focus and optical centre a virtual, magnified and erect image is formed



$$\text{Magnifying power (MP)} = \frac{\text{visual angle with instrument } (\beta)}{\text{maximum visual angle for unaided eye } (\alpha)} = \frac{\frac{h}{-u}}{\frac{h}{-D}} = \frac{D}{u}$$

(i) When the image is formed at infinity :

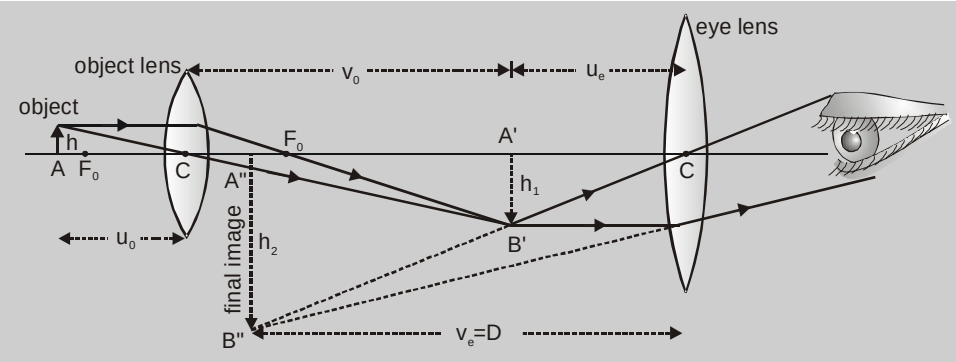
$$MP = \frac{D}{u} = \frac{D}{f}$$

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(ii) If the image is at minimum distance of clear vision D :

$$MP = \frac{D}{u} = 1 + \frac{D}{f}$$

(b) Compound Microscope



Compound microscope is used to get more magnified image. In this condition, the final image is virtual, invertred and magnified.

Total magnifying power = Linear magnification × angular magnification

$$MP = m_0 m_e = \frac{v_0}{u_0}\frac{D}{u_e}$$

When final image is formed at minimum, distance of distinct vision.

$$MP = \frac{v_0}{u_0}\left[1 + \frac{D}{f_e}\right]$$

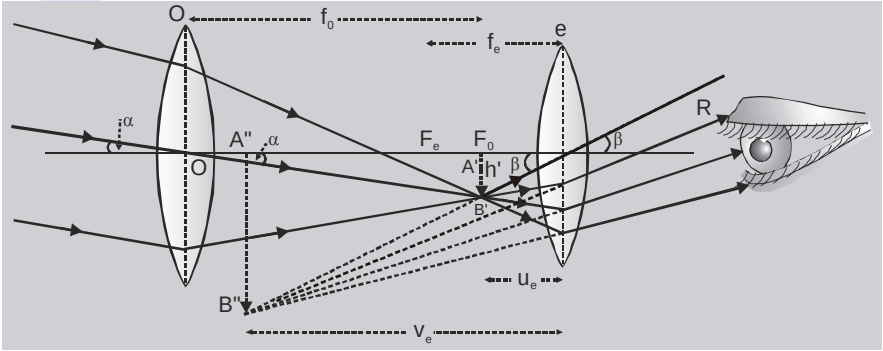
Length of the tube = $v_0 + |u_e|$

When final image is formed at infinity "

$$MP = \frac{v_0}{u_0}\left[\frac{D}{f_e}\right]$$

Length of the tube $L = v_0 + f_e$

(c) Astronomical Telescope



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A telescope is used to see distant object, objective lens forms the image A'B' at its focus. This image A'B' acts as a object for eye-piece and it forms final image A"B".

$$MP = \frac{\text{visual angle with instrument } (\beta)}{\text{visual angle for unaided eye } (\alpha)} \Rightarrow MP = -\frac{f_0}{u_e}$$

(i) If the final image is at infinity

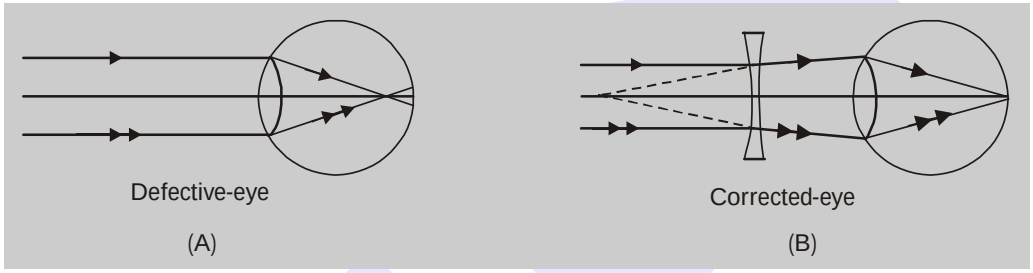
so $MP = -\frac{f_0}{f_e}$ and lenght of the tube $L = f_0 + f_e$

(ii) If the final image is at D

So $MP = -\frac{f_0}{u_e} = -\frac{f_0}{f_e}\left[1+\frac{f_e}{D}\right]$ Lenght of the tube is $L = f_0 + u_e$

30. DEFECTS OF EYES

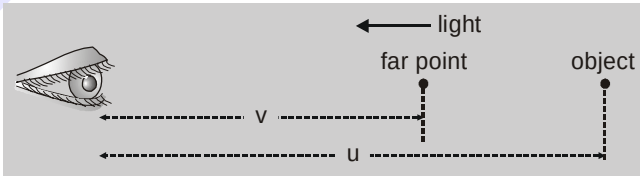
(a) Myopia [or Short-sightedness or Near- sightedness]



(i) Distant object are not clearly visible, but near object are clearly visible because image is formed before the ratina.

(ii) To remove the defect concave lens is used.

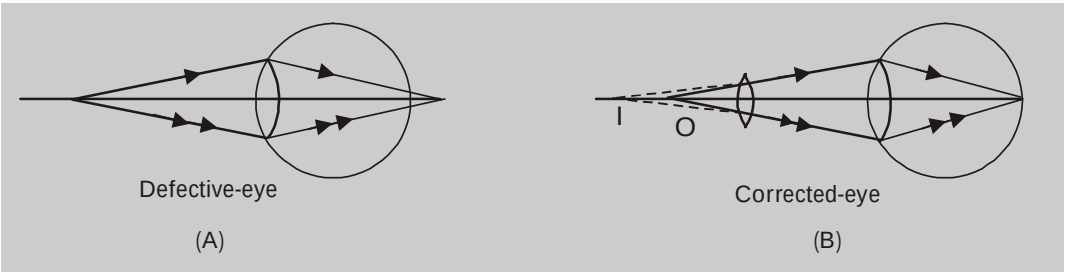
◎ The maximum distance. Which a person can see without help of spectacles is known as far point.



◎ In this case image of the object is formed at the far point of person.

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}=P \quad \Rightarrow \quad p = -ve$$

(b) Long-sightedness or Hypermetropia



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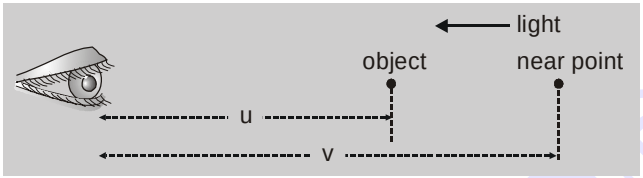
(i) Near object are not clearly visible but far object are clearly visible.

(ii) The image of near object is formed behind the retina.

(iii) To remove this defect convex lens is used.

Near Point :-

The minimum distance which a person can see without help of spectacles.



- In this case image of the object is formed at the near point.
- If reference of object is not given it is taken as 25 cm.

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}=P \quad \Rightarrow \quad p = +ve$$

(c) Presbyopia

In this case both near and far object are not clearly visible.

To remove this defect two separate spectacles one for myopia and other for hypermetropia are used or bifocal lenses are used.

(d) Astigmatism

This defect arises due to imperfection of complete spherical shape of eye lens. In this defect eye can not see object in two orthogonal direction clearly. It can be removed by using cylinderical lens in perticular direction.

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